

Open-vocabulary Object Detection via Vision and Language Knowledge Distillation

Xiuye Gu, Tsung-Yi Lin, Weicheng Kuo, Yin Cui

Google Research and Nvidia

ICLR 2022 — Beamer summary

- Slides summarize the camera-ready paper and preserve core math, figures, and results.
- Code and demo: <https://github.com/tensorflow/tpu/tree/master/models/official/detection/projects/vild>

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Related Work

- Zero-shot recognition:** attributes, class hierarchies, and vision-language embeddings (DeViSE, CLIP, ALIGN).
- Open-vocabulary classification:** Contrastive learning on large image-text corpora enables **zero-shot** transfer.
- Object detection at scale:** costly to expand vocabularies with instance labels; zero-shot detection aligns regions to text.
- Our approach: **distill** a strong image-text teacher (CLIP/ALIGN) into a **two-stage detector** for open-vocabulary detection at object level.

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Motivation

- Goal: **open-vocabulary object detection** — detect objects from **arbitrary text** inputs (beyond training labels).
- Challenge: detection datasets are long-tailed (Zipf's law); scaling labeled categories is **expensive**.
- Opportunity: Internet-scale **image-text** pairs (e.g., CLIP, ALIGN) offer rich supervision.
- Idea: Distill knowledge from a **pretrained open-vocabulary classifier** into a fast two-stage detector (**ViLD**).

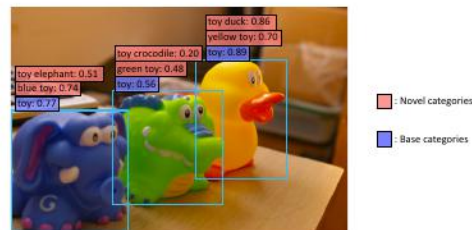
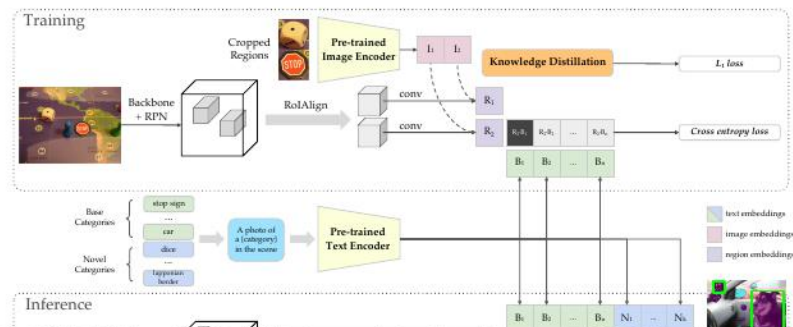


Figure: Teaser: After training on base categories (purple), detect **novel** categories (pink) from text.

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Method Overview: ViLD

- Teacher: Pretrained **open-vocabulary classifier** with image encoder $\mathcal{V}(\cdot)$ and text encoder $\mathcal{T}(\cdot)$.
- Student: Two-stage detector producing region embeddings $\mathcal{R}(\phi(I), r)$.
- Two complementary signals:
 - ViLD-text:** classify regions using **fixed text embeddings** from $\mathcal{T}$.
 - ViLD-image:** **distill** region embeddings to match $\mathcal{V}$ proposal embeddings.



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Method I: Localization for Novel Categories

- Modify two-stage detector (e.g., Mask R-CNN) to use **class-agnostic** localization:
 - 2nd-stage box regression: **one** box for all categories.
 - Mask head: **one** mask head for all categories.
- Trained on **base** categories, but generalizes to **novel** objects.
- RPN proposals remain strong on novel classes (small AR drop vs. base+novel training).

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Method II: Open-vocabulary Detection with Cropped Regions

- Use RPN proposals $\tilde{r} \in \tilde{P}$, crop and resize to feed **teacher image encoder** $\mathcal{V}$.
- Image embeddings** (context ensemble of $1\times$ and $1.5\times$ crops):

$$\mathcal{V}(\text{crop}(I, \tilde{r}_{\{1\times, 1.5\times\}})) = \frac{\mathbf{v}}{\|\mathbf{v}\|}, \mathbf{v} = \mathcal{V}(\text{crop}(I, \tilde{r}_{1\times})) + \mathcal{V}(\text{crop}(I, \tilde{r}_{1.5\times})).$$
- Text embeddings**: prompt-ensemble using $\mathcal{T}$ on category names/synonyms.
- Score by cosine similarity + softmax; per-class NMS for final detections.
- Strong on novel categories but **slow** (R-CNN style).

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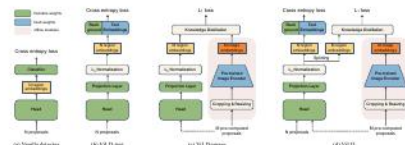
Method III: ViLD-text and ViLD-image (Core)

ViLD-text

- Replace classifier with fixed **text** embeddings $\mathcal{T}(C_B)$ plus learnable **background** e_{bg} .
- Train with **cosine** similarity and **softmax** at temperature τ :

$$\mathcal{L}_{\text{ViLD-text}} = \frac{1}{N} \sum_{r \in P} \mathcal{L}_{CE}(\text{softmax}(\mathbf{z}(r)/\tau), \text{gt})$$

Figure: Architecture and objectives



ViLD-image

- Distill **image** embeddings using offline proposals $\tilde{r} \in \tilde{P}$:

$$\mathcal{L}_{\text{ViLD-image}} = \frac{1}{M} \sum_{\tilde{r} \in \tilde{P}} \|\mathcal{V}(\text{crop}(I, \tilde{r}_{\{1\times, 1.5\times\}})) - \mathcal{R}(\phi(I), \tilde{r})\|_1.$$

Total loss

$$\mathcal{L}_{\text{ViLD}} = \mathcal{L}_{\text{ViLD-text}} + w \cdot \mathcal{L}_{\text{ViLD-image}}.$$

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Experimental Method and Setting

Experimental Method (design and process)

Benchmarks:

- LVIS v1: 866 **base** + 337 **rare** (novel) categories; primary metric: AP_r (rare).
- COCO split: 48 base / 17 novel (generalized setting).

Protocol:

- Train only on base category annotations; **no** novel-instance labels.
- Use teacher (CLIP/ALIGN) to produce text/image embeddings for ViLD training.

Experimental Setting (datasets, parameters, environment)

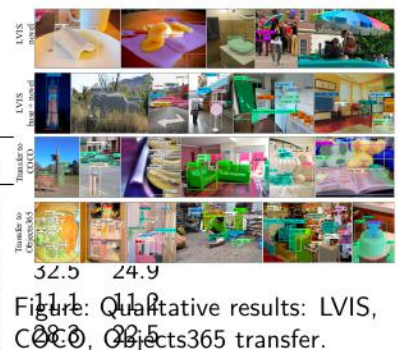
- Detector: Mask R-CNN with **ResNet-50 FPN**, 1024×1024 , large-scale jitter $[0.1, 2.0]$.
- Opt: batch 256 (sync BN), WD $4e-5$, LR 0.32, 180k iters with LR drops at 0.9/0.95/0.975.
- Teacher: **CLIP** ViT-B/32 (224×224); also **ALIGN**/CLIP ViT-L/14

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Table: Main mask AP on LVIS (ResNet-50 unless noted).

Method	AP _r	AP _c	AP _f	AP
Supervised-RFS (base+novel)	12.3	24.3		
GloVe baseline	3.0	20.1		
ViLD-text	10.1	23.9		
ViLD-image	11.2	11.3		
ViLD ($w=0.5$)	16.1	20.0		
ViLD-ensemble ($w=0.5$)	16.6	24.6		
ViLD-ensemble + ALIGN	26.3	27.2	32.9	29.3



As shown in Table 1, **ViLD** surpasses supervised on novel categories: **ALIGN**

Deficiencies and Future Research

- Deficiencies
- **R-CNN** variant is slow at inference (per-proposal cropping).
 - Competition between **ViLD-text** and **ViLD-image** objectives.
 - **Localization** quality occasionally lags (e.g., masks by low-level cues).
 - Teacher limitations: aspect-ratio distortion, multiple objects per crop.

- Future Research
- Stronger teachers and **better proposal** networks for unseen classes.
 - Multi-task schedules to **reduce contention** between text and image losses.
 - Improved **region-aware** vision-language pretraining (object-level grounding).
 - Systematic **vocabulary expansion** to attributes / fine-grained classes.



Table: Effect of distillation weight w and loss on LVIS (mask AP).

Distill setting	AP _r	AP _c	AP _f	AP
No distill ($w=0.0$)	10.4	22.9	31.3	24.0
$\mathcal{L}_2, w=0.5$	13.7	21.7	31.2	24.0
$\mathcal{L}_1, w=0.1$	14.0	20.9	31.2	23.8
$\mathcal{L}_1, w=0.5$	16.3	19.2	27.3	21.9
$\mathcal{L}_1, w=1.0$	17.3	18.2	25.1	20.7

- $\mathcal{L}_1$ distillation improves AP_r more than $\mathcal{L}_2$, but trades off AP_{c,f}.
- Prompt ensembling for text improves AP_r slightly (16.1 vs. 15.7).

Thank you

- Paper: Open-vocabulary Object Detection via Vision and Language Knowledge Distillation (ICLR 2022)
- Code and demo: <https://github.com/tensorflow/tpu/tree/master/models/official/detection/projects/vild>
- Contact: xiuyegu, weicheng, yincui@google.com; tsungyil@nvidia.com



Figure: LVIS categories.



Figure: Fine-grained bird species (CUB-200-2011).